

ShiftShield

Who does what: Sameer and Gufran

A working guide for a team of two. It covers the two days before kickoff, each of the four hackathon days, the points where your work meets, and what to do when something slips. Prepared October 6, 2026. Read it together with ShiftShield_Final_Plan_v7.pdf, which has the full design.

	Sameer	Gufran
Role	Engine, rulebook and evidence	Cloud and product
In one line	Everything that calculates or reads	Everything that runs or is seen
Owns	Heat estimate, work-rest schedule, rulebook agent, replay and results, README, video, writeup, blog	AWS account and deploy, database, alert logic and email, API, dashboard, public heat check, worker QR page, rest record, demo link
Main language	Python	Python for Lambda; JavaScript for the pages
Outside contacts	HOD and any expert reviewer	Supervisor, workers and the field-test site
Final say on	Scope and cuts (as team leader), numbers and claims	What is deployed and when the build is frozen

If your strengths are the other way round

This split assumes Sameer is more comfortable with Python and data, and Gufran with AWS and web pages. If it is the reverse, swap the two columns completely. Do not split one role between two people; each part needs one owner who can say it is done.

The rule that protects you from disqualification

Until the hackathon opens on Thursday, October 8, nobody creates the repository or writes project code. Research, reading, account setup, tool installation, interviews and practice on AWS workshops are all allowed. Everything in Section 2 is that kind of work.

SECTION 1

How the product fits together, and who owns each part

Seven parts. Each has one owner. The arrows show the order information moves in.



Orange boxes are Gufran's; the others are Sameer's.

Part	Owner	In plain words	Depends on
Forecast fetch	Sameer	Gets hourly temperature, humidity, wind and sunlight for the site's location from Open-Meteo.	Nothing
Heat estimate (WBGT)	Sameer	Turns the forecast into a heat-stress number every 15 minutes, adjusted for the site's sun and shade, with a safety margin.	Forecast
Schedule	Sameer	Looks the number up in a fixed table copied from NIOSH and returns a band with work minutes, rest minutes and water.	Heat estimate, NIOSH table
Rulebook agent	Sameer	Reads a heat action plan, pulls out each worker obligation with its sentence and page, rejects any quote not found on that page.	Bedrock access, the two plans
Daily compliance plan	Sameer	For each hour, shows what the plan requires and what the schedule adds, and applies the stricter.	Schedule, approved rulebook
AWS stack and deploy	Gufran	The SAM template that creates the Lambdas, tables, schedule and hosting with one command.	AWS account
Alert logic, log, email	Gufran	Every 15 minutes: call the engine, decide whether to alert, send once, write to the log.	Engine output shape
API	Gufran	The routes the pages call: site, forecast, plan, rest record, rulebook approval, compliance.	Engine, tables
Dashboard and approval screen	Gufran	What the supervisor and safety officer see, including accept, edit or reject for each extracted rule.	API, rulebook candidates
Worker QR page and rest record	Gufran	The page behind the QR code with break and symptom buttons, and the four states of each alert.	API
Public heat check and demo link	Gufran	No-login advice for any location; a link that plays a replayed hot day by itself.	Engine, replay data
Replay, results, README, video	Sameer	Runs a past hot day through the system, produces the numbers, writes and records the story.	Everything above

SECTION 2

Before kickoff: Tuesday October 6 and Wednesday October 7

Research and setup only. Every item here saves hours during the four days.

Sameer

Task	What to do, step by step	Done when
NIOSH heat tables	Open the NIOSH 2016 document "Occupational Exposure to Heat and Hot Environments" (free on the CDC website). Find the recommended limits by work rate and acclimatisation, and the work-rest guidance. Write down the page number of every table you will copy. Do not type the numbers into code yet; note them on paper or in a document.	You can point to the exact page for every band limit and every work-rest value.
Obligation list for two plans	Download Delhi's Heat Wave Action Plan 2026 and your own state's or city's plan. Check you can select text in the PDF. Read each one and fill a sheet with these columns: plan name, page, exact sentence, who must act, what must be done, when it applies, type (rest, water, shade, hours, first aid, other). Include only obligations that affect workers at a site.	One sheet per plan, with every row carrying a page number and a sentence copied word for word. This is the answer key the agent is scored against.
Site factor sources	For shade, wind shelter and built-up surroundings, find one published source each that gives a number or a range. Search for urban heat island studies of Indian cities and WBGT estimation papers. Keep the title, link and page. Remember the field study found surface type matters little, so spend your time on sun and shade.	A notes file with one source and one value or range per factor.
Pick the replay day	Choose your pilot city. Look for a day in April to June with a reported heatwave. Open the Open-Meteo historical weather page and confirm hourly temperature, humidity, wind and solar radiation exist for that date and place. Also pick one mild day as a control.	Two dates and one location written down, with data confirmed.
HOD review	Show ShiftShield_Project_Proposal_v3.pptx to your HOD. Ask two things: is the rulebook and rest-record direction acceptable under the problem statement, and can they suggest someone in occupational health or safety to look at the threshold table.	You have a yes on the direction, and a name or a no for the reviewer.

Gufran

Task	What to do, step by step	Done when
AWS account	Create a new AWS account (a debit card or RuPay works; expect a verification charge of about two rupees). Turn on multi-factor sign-in for the root user. Create one admin user for each of you and stop using root. In Billing, create a budget with an email alert at a low amount. Choose one region and write it down; use it for everything.	Both of you can sign in as your own user, and a budget alert exists.
Bedrock model access	In the Bedrock console, in your chosen region, request access to one text model and one model that can read documents or images. Approval can take time, which is why this is done now.	The console shows access granted for both models.
Tools on both laptops	Install Python 3.12, Node 20, Git, the AWS CLI and the SAM CLI. Run aws configure with your own user's keys. Run sam --version and aws sts get-caller-identity to confirm both work.	Both laptops print a version and your account identity without errors.
Practice	On AWS Builder Center, do one beginner workshop that covers Lambda and DynamoDB, in a sandbox. This is practice in a throwaway environment, not project work, so it is allowed.	You have deployed and invoked one Lambda that writes to a table.
Interviews	Speak to one site supervisor and two or three workers; a campus construction site is fine. Ask the supervisor: what do you do when a heatwave SMS arrives, who decides breaks, could you start earlier, what phone and language do you use. Ask the workers: do you have a smartphone, would you scan a code to say you got a break, what would stop you, is there water and shade. Ask permission to quote them, and to film on Saturday.	Written notes, one usable quote, and a clear yes or no on whether workers would scan.
Field-test site	Agree a place and a time on Saturday, October 10, where three or more people can scan the QR code during a replayed break.	A named contact and a time.

Both of you

- Finish student verification on AWS Builder Center. Prizes and the interview shortlist wait on it.
- Check the hackathon schedule page each evening; the kickoff and deadline hours were not published when this was written.
- Join the WeMakeDevs Discord and Builder Center space.
- If Gufran's interviews say workers will not scan a QR code, tell Sameer the same day. That changes the design and is far cheaper to fix now than on Saturday.

SECTION 3

The four days, person by person

Times are given as parts of the day because the official hours are not published yet. Shift them once they are.

Thursday, October 8: the engine and the skeleton

When	Sameer	Gufran
Kickoff	Join the kickoff. Agree the engine output shape with Gufran (Section 4) before either of you writes code.	Create the public repository after the clock starts. Add a README stub, licence and a credits file. Give Sameer access.
First half	Forecast fetch from Open-Meteo. WBGT calculation using an open-source library. Site adjustment for sun and shade. First unit tests.	SAM template with one Lambda, the DynamoDB tables and an HTTP API. First deploy. Budget check.
Second half	Threshold table file transcribed from NIOSH with page references. Schedule lookup. Tests that the same input always gives the same schedule.	Package Sameer's WBGT library into a Lambda layer and confirm it imports. This is the riskiest step of the day; do it early.
Evening	Commit the hand-made obligation lists as test data. Produce a chart of the heat estimate for two site profiles.	Evaluation Lambda calls the engine with a fixed site and stores the result. Frontend project created on Amplify.

End-of-day check: the engine passes its tests on Sameer's laptop and runs inside a Lambda on AWS; both of you have commits in the history.

Friday, October 9: alerts and the rulebook

When	Sameer	Gufran
First half	Rulebook agent: send plan pages to Bedrock, get candidate obligations with quote and page, reject any quote not found on that page.	EventBridge schedule every 15 minutes. Alert trigger with no duplicates. Append-only log. Email through SNS with an acknowledge link.
Second half	Daily compliance plan: merge approved rules with the schedule, stricter wins. Score the agent against the answer key and fix the prompt.	API routes. Dashboard showing the forecast chart, the current plan and the log. Public heat check page.
Evening	Replay mode: run the chosen hot day through the same engine and trigger. Hand Gufran the replay data for the demo link.	Approval screen for the rulebook: accept, edit or reject each rule, showing its sentence and page.

End-of-day check: a scheduled run on AWS sends one correct email; one heat action plan has gone from PDF to approved rules; the replay produces a full day of bands.

Saturday, October 10: the loop, the evidence, the freeze

When	Sameer	Gufran
First half	Run the replay for both site profiles and the control day. Fill the results table. Run the sensitivity check.	Worker QR page with break and symptom buttons. Rest record with its four states on the dashboard. Self-playing demo link.
Midday	Join the field test and film it. Collect one line from the supervisor or a worker.	Field test: three or more people scan the code during a replayed break. Fix what confuses them.
2 PM checkpoint	Together: is the core live (alert, rest record, rulebook, public page)? If not, both stop everything else.	Same.
Second half	README in the winners' structure. Video script with the measured numbers. Architecture diagram.	Bonus features in order, only if the core passed: escalation chain, voice alert, certificate, shade payback. CloudWatch alarm.
9 PM	Freeze. Record the product parts of the video while everything works.	Freeze. No new features after this; fixes only.

End-of-day check: a stranger can open the live link and complete the whole flow; the numbers for the video exist.

Sunday, October 11: submit

When	Sameer	Gufran
Early	Record the voice, edit to under three minutes, upload to YouTube as public or unlisted.	Open every link signed out, on a phone. Warm the Lambdas. Confirm the repository is public and has no secrets.
After the form opens	Finish the writeup: problem, build, where AWS fits, AI tools used, limits. Hand the text to Gufran.	Fill and submit the form. Take a screenshot of the confirmation.
After submitting	Publish the blog on AWS Builder Center and add its link if the form allows edits.	Keep the deployment running until judging ends. Post the project in the Discord.

End-of-day check: submission confirmed, well before the deadline.

SECTION 4

Where your work meets

Three agreements to make on Thursday morning. They let you build in parallel without waiting on each other.

Agreement 1: what the engine returns

Gufran builds the alert logic and the pages against this shape while Sameer is still writing the engine. Agree the field names once and do not change them without telling each other. The values below are placeholders; real work and rest minutes come from the NIOSH table.

```
{
  "site_id": "site-001",
  "time": "2026-05-21T13:15:00+05:30",
  "wbgt_c": 31.4,
  "margin_c": 1.0,
  "band": "C",
  "band_label": "High",
  "work_min": "<from table>",
  "rest_min": "<from table>",
  "water": "<from table>",
  "next_change": { "time": "2026-05-21T14:00:00+05:30", "to_band": "D" },
  "forecast_age_min": 20,
  "engine_version": "0.1",
  "thresholds_version": "v1"
}
```

Agreement 2: what a rule looks like

Sameer's agent produces candidates in this shape; Gufran's approval screen shows them and stores the decision.

```
{
  "rule_id": "delhi-2026-007", "plan": "Delhi Heat Wave Action Plan 2026",
  "type": "rest_window", "applies_when": "<condition from the plan>",
  "requirement": "<what must be done>",
  "quote": "<exact sentence>", "page": 14,
  "status": "candidate" // candidate, approved, rejected
}
```

Agreement 3: who owns which file

Folder	Owner	The other person
engine/ (forecast, WBGT, schedule, thresholds file, tests)	Sameer	Reads it, does not edit it. Asks for changes.
rulebook/ (agent, quote check, compliance plan, answer keys)	Sameer	Same.
infra/ (SAM template), functions/ (Lambdas), web/ (pages)	Gufran	Same.
README, docs/, video script	Sameer	Gufran supplies the architecture and deploy steps.

Staying out of each other's folders is the simplest way for two people to avoid merge conflicts under time pressure.

SECTION 5

How you work together

Habit	How	Why
Commits	Small, every hour or so, straight to main, with a short message saying what changed. Pull before you push.	The judges check that the history matches the event dates. Winners last time had 35 to 87 commits.
Check-ins	Ten minutes, twice a day: what I finished, what I am doing next, what is blocking me.	A blocker found at noon costs an hour; found at midnight it costs the night.
AI tools log	Keep one file listing every AI coding tool either of you uses, as you use it.	The rules require the list in the writeup.
Credits file	Add every library, API, dataset and asset the day you start using it.	Missing credit is treated as copying.
Secrets	No keys in the repository, ever. Use environment variables and Secrets Manager.	The repository is public.
Disagreements	Sameer decides scope and claims; Gufran decides what is safe to deploy. Decide in five minutes and move on.	There is no time to debate.
Sleep	Both of you sleep on Friday and Saturday night.	Tired people break working code.

The Saturday 2 PM rule

At 2 PM on Saturday, check four things on the live link: one correct alert arrives, the rest record changes state when the supervisor taps and a worker scans, one heat action plan shows approved rules in the daily plan, and the public heat check gives advice. If any one fails, both of you stop all other work and fix it. Bonus features are not started until all four pass.

If something slips, cut in this order

Order	Cut	Who decides	What you show instead
1	All four bonus features	Sameer	Named as next steps in the README.
2	AI-written alert wording	Sameer	Fixed message templates.
3	Second heat action plan	Sameer	One plan end to end; say "verified on one".
4	Live extraction on demo day	Sameer	The approved rulebook shipped as data, extraction shown on one short section.
5	Symptom buttons	Gufran	Break confirmation only.
Never	Engine, alert, rest record, daily compliance plan, public page, replay, video	Nobody	These are the product.

SECTION 6

Bonus features and the submission, with owners

Order	Bonus feature	Sameer's part	Gufran's part
1	Escalation chain	Decide the timings and who is in the chain.	Step Functions workflow that waits for the acknowledgement and times out to the next person.
2	Voice alert	Check which Indian languages Polly supports; write the spoken text.	Polly call and an audio link in the alert.
3	Heat-day certificate	Define the figures on it and compute them from the log.	KMS signing, the certificate page and the verify page.
4	Shade payback	Re-run the engine with shade on and compute minutes and rupees.	A panel on the dashboard.

Take them strictly in this order. One finished bonus feature is worth more than four started.

Submission checklist

Item	Owner	Check	Done
Public repository	Gufran	History runs October 8 to 11; no secrets; setup steps work on a clean machine.	☐
README	Sameer	Problem as a human moment, who it is for, try it, where AWS fits, safety model, tests, limits, AI tools, credits.	☐
Live link and demo link	Gufran	Open signed out, on a phone, on mobile data.	☐
Demo video	Sameer	Under three minutes; shows the working system and the AWS console; replay is labelled.	☐
Writeup	Sameer	Problem, build, where AWS fits, AI tools used, limitations.	☐
Form submitted	Gufran	Confirmation screenshot saved; both names and details match Builder Center.	☐
Blog on Builder Center	Sameer	Published and linked.	☐
Deployment kept alive	Gufran	Until results are announced.	☐

If one of you is unavailable for a day

The other person works only on the "Never cut" list in Section 5, in the order the information flows: engine, alert, rest record, compliance plan, public page. Everything else waits. Tell each other as early as possible.

What to tell each other tonight

Sameer: which pilot city and which state's plan you are choosing. Gufran: which AWS region you are using and when the interviews are. Then each of you starts on the first row of your table in Section 2.